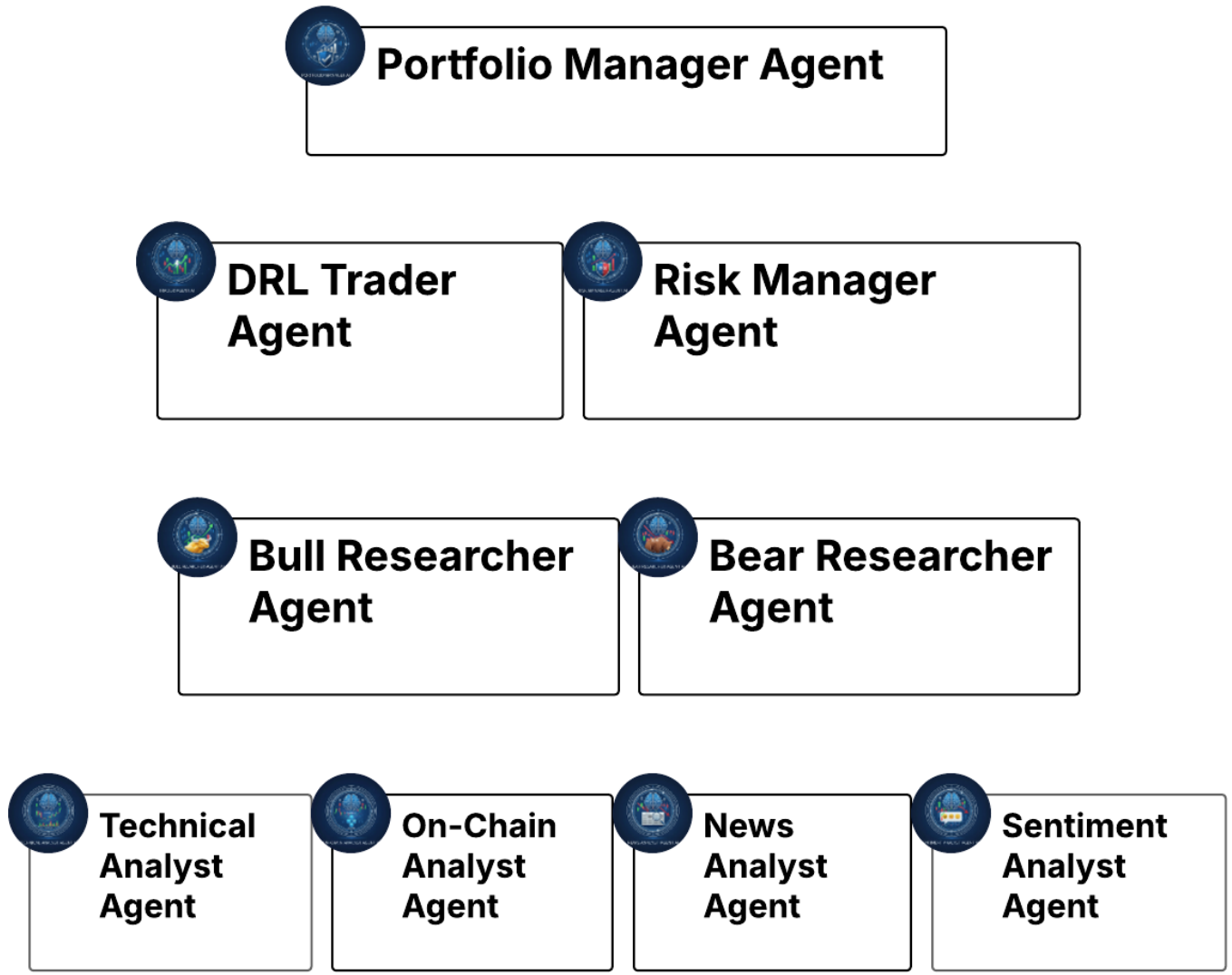
Multi Agent System (MAS) for Crypto Portfolio Management

Hammad Qaiser

According to Modern Portfolio Theory (MPT)

All investors should be **rational**.

MAS Architecture



Multi-Agent Orchestration and Collaboration Framework

LangGraph

LangChain

TradingAgents Framework

TAURIC RESEARCH

TradingAgents: Multi-Agents LLM Financial Trading Framework

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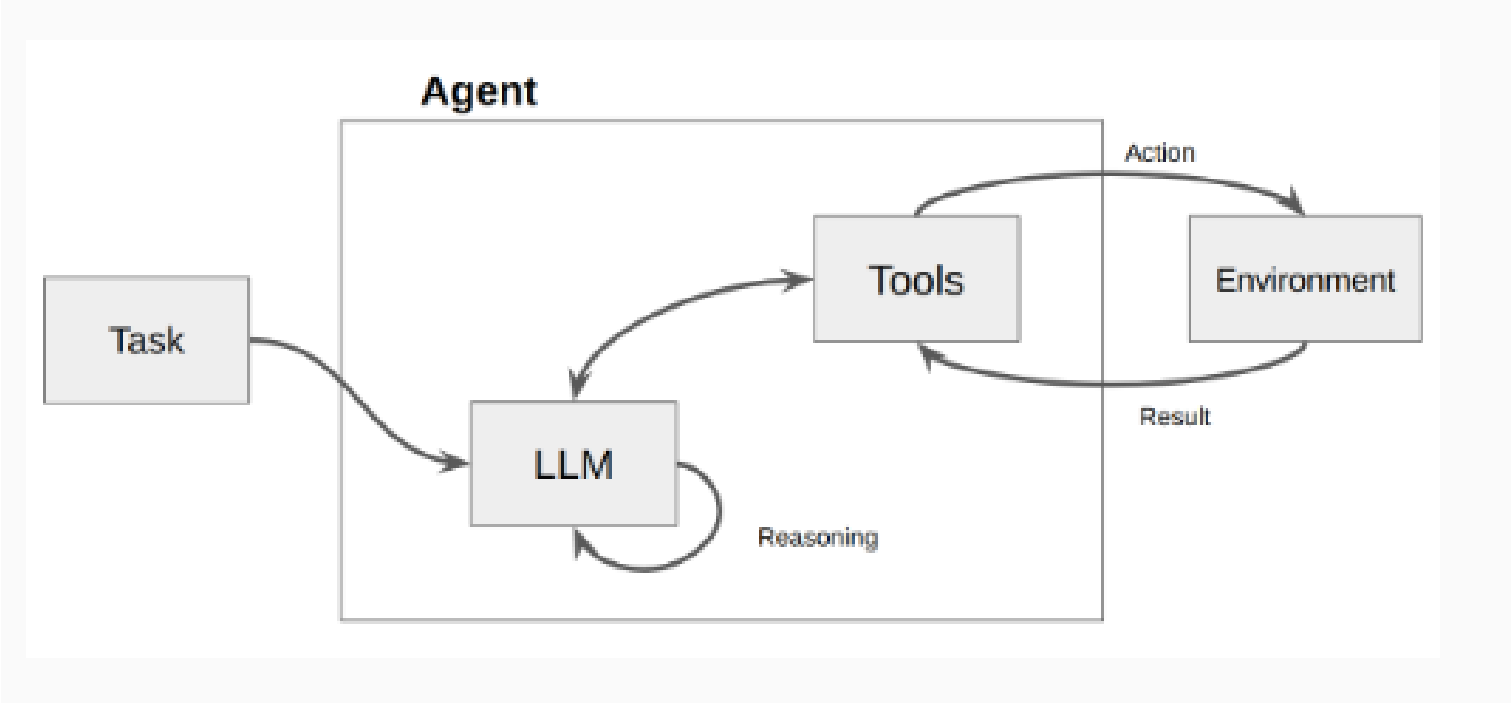
ReAct Framwork



Common Goal of MAS

Linear regression equation

$$\hat{Y} = bX + a$$

MAS Objective

- **Y (Asset Return):** The asset return
- **X (Market Return):** The return of the overall crypto market index
- **b (Beta):** The asset's sensitivity to market movements.
- **a (Alpha):** The **Excess Return** generated by the MAS.

MAS will focus on Alpha:

- Simple Buy-and-Hold strategy can achieve **Beta** returns.
- Primary goal of the MAS to consistently generate **positive Alpha**. This indicates superior performance relative to the risk taken, maximizing the **Sharpe Ratio**.

Key Performance Indicators (KPIs)

- **Sharpe Ratio:** Excess return relative to total volatility.
- **Maximum Drawdown (MDD):** Largest percentage loss from a peak.
- **Calmar Ratio:** Annualized return divided by MDD.
- **Win Rate:** Percentage of profitable trades.